

## 10-DAY SPRINT PLAN

# Vesper

Frontend and backend plan for a working demo of one Mumbai resort

**Goal:** a working end-to-end demo of Vesper for one fictional Mumbai beach resort (modelled on public data from a 145-room Madh Island resort). It combines the new modules (staff app, guest QR, inventory, front desk) with every AI feature from the HackCelestial prototype.

**Reality check:** 10 days delivers a **demo-quality MVP**. Most AI logic already exists in the repo, so Days 6–8 focus on reconnecting, fixing and restyling it.

### Plan at a glance

| Day | Theme                   | Main outcome                                                       |
|-----|-------------------------|--------------------------------------------------------------------|
| 1   | Foundation              | Setup, Vesper design system, database, login, prototype bug fixes  |
| 2   | Roles & demo resort     | Permissions, admin panel, Mumbai demo data, PMS/BMS connectors     |
| 3   | Staff app               | Attendance, tasks, housekeeping room board                         |
| 4   | Guest QR & requests     | Room QR ordering, staff issue and shortage requests                |
| 5   | Inventory & front desk  | Stock tracking, bookings, guest visit tracking                     |
| 6   | Action queue & revenue  | Full action card flow, forecast and pricing page                   |
| 7   | Maintenance & workforce | Risk and sensor screens, auto roster, staffing gaps                |
| 8   | Guests & AI concierge   | Guest DNA, sentiment, retention offers, concierge chat             |
| 9   | Dashboard & system      | Owner dashboard, simulator, learning, shadow mode, import, 3D twin |
| 10  | Polish & demo           | Mobile testing, day simulator, tests, deployment                   |

### Four surfaces, one backend

- **Vesper Admin** (website): owner, GM and managers.
- **Vesper Staff** (installable PWA): attendance, tasks, requests, QR scanning.
- **Vesper Guest** (no login): opened by scanning the room QR code.
- **Marketing site**: backlog, after the demo works.

# Daily Plan

## DAY 1

### Foundation

Everyone builds on the same base

#### FRONTEND

- Next.js admin site and PWA staff app shells
- Vesper theme: sage/sand/gold tokens, serif + Inter, Lucide icons
- shadcn/ui base: Button, Input, Card, Badge, Skeleton, Toast
- Login screens; app shell with sidebar and top bar

#### BACKEND

- Restructure FastAPI into modules
- PostgreSQL schema: property, departments, users, rooms
- Secure login; protect all routes and WebSocket
- Fix prototype bugs: per-date rates, safe undo, timezone, sentiment

✓ **End of day:** Team logs in to both apps; theme and database ready; known bugs fixed.

## DAY 2

### Roles & Demo Resort

The right people see the right things

#### FRONTEND

- Admin panel: users, roles, permission matrix editor
- Role-based sidebar and bottom nav
- Access-denied, empty-state and 404 pages
- Resort settings page

#### BACKEND

- Permission system (e.g. rates:approve, attendance:mark)
- Seed demo resort: 145 rooms, outlets, ~180 staff, equipment
- Demo PMS (bookings) and BMS (sensor) connectors
- Audit log for every decision

✓ **End of day:** Log in as Admin, GM, manager and employee; each sees a correct view.

## DAY 3

### Staff App

Staff do their daily work from a phone

#### FRONTEND

- Attendance: QR or location check-in/out
- My Tasks: cards, filter chips, swipe to complete
- Housekeeping room board: dirty → cleaning → ready
- Manager view: team attendance and task progress

#### BACKEND

- Attendance APIs with shift and location checks
- Task APIs: create, assign by department, update
- Room status APIs linked to check-outs
- WebSocket events for live updates

✓ **End of day:** A housekeeper checks in and cleans a room; the manager sees it live.

## DAY 4

### Guest QR & Staff Requests

Guests and staff raise requests instantly

#### FRONTEND

- Guest page: room service menu, cart, cleaning, towels, issue
- Live request tracker and 1-tap rating
- Staff app: requests inbox with accept and timer
- Staff app: report broken item or shortage with photo

#### BACKEND

- Room QR tokens valid only while occupied
- Request routing by department with SLA timers
- Overdue alerts to managers
- Issue reports with photo upload; merge duplicates

✓ **End of day:** Scan a room QR, order food, kitchen accepts, guest sees status.

## DAY 5

# Inventory & Front Desk

Stock and guests tracked in one place

## FRONTEND

- Inventory: food, linen, beds, toiletries, parts
- Low stock and expiry badges; stock in/out modal
- Front desk: room board, bookings, check-in/out drawer
- Guest profile with visit history

## BACKEND

- Inventory APIs: stock levels, minimums, expiry
- Auto-deduct stock on completed room service
- Booking and check-in APIs on demo PMS data
- Visit tracking from stays, restaurant, spa, QR orders

✓ **End of day:** A room service order lowers stock; low stock raises an alert.

## DAY 6

# Action Queue & Revenue

AI suggestions managers can trust and act on

## FRONTEND

- Action card: drivers, confidence bar, ₹ impact, urgency
- Approve, adjust modal, snooze menu, dismiss reasons
- Undo countdown toast
- Revenue page: 30-day forecast with confidence band, rate card, competitor table

## BACKEND

- Reconnect demand engine (Prophet + XGBoost) to new schema
- Per-date rate execution with safe revert
- Card ranking, claim lock, permission checks per engine
- Purchase suggestions from low stock

✓ **End of day:** Manager adjusts and approves a price for one date, then undoes it safely.

## DAY 7

# Maintenance & Workforce

Fix things before they break; staff where needed

## FRONTEND

- Asset list and health cards with risk gauge
- Sensor trend chart with anomaly markers; service window
- Work order form and QR code generator for machines
- Roster grid, staffing need vs scheduled chart, gap warnings

## BACKEND

- Reconnect maintenance engine to BMS data and QR logs
- Work orders created from approved cards
- Reconnect workforce engine (OR-Tools) to attendance data
- Auto roster API with leave and shift rules

✓ **End of day:** A chiller is flagged, repair booked on a quiet day, roster adjusts.

## DAY 8

# Guests & AI Concierge

Know every guest and keep them coming back

## FRONTEND

- Guest DNA card with preference chips
- Department sentiment trend chart
- Retention: at-risk guests and offer composer
- Concierge chat: bubbles, typing state, sources, escalate button

## BACKEND

- Replace keyword sentiment with a real model
- Guest DNA from stays, orders and ratings
- At-risk detection when visits drop; offer cards
- Concierge (RAG + Claude) with auth and rate limits

✓ **End of day:** Guest asks the concierge a question; an at-risk guest gets an approved offer.

**DAY 9**

## Dashboard & System Features

The owner sees everything; the system is transparent

**FRONTEND**

- Owner dashboard: KPI tiles, charts, alerts, live feed
- Simulator: price, staffing and promo sliders
- Learning page, cold-start banner, shadow mode toggle
- CSV import with dry-run errors; outbox; 3D resort twin (lazy-loaded)

**BACKEND**

- Dashboard summary APIs
- Simulator, readiness and learning endpoints on new data
- Shadow mode across all executors
- Notification outbox with retry (mock WhatsApp/SMS/email)

✓ **End of day:** Owner runs a what-if, toggles shadow mode, and sees engine accuracy.

**DAY 10**

## Polish, Test & Demo

A smooth demo anyone can run

**FRONTEND**

- Test on real phones; fix layouts
- Skeletons, empty states, consistent spacing
- Remove inline styles on key screens
- Demo script and screenshots

**BACKEND**

- Day simulator: plays out a live resort day
- Tests for permissions, bugs fixed and key flows
- Deploy: Vercel (web), Railway (API, database)
- Fresh demo data on deploy

✓ **End of day:** Full demo from guest QR to AI action to owner dashboard, online.

# Prototype Features Coverage

Every feature from the HackCelestial repo and where it is built:

| Prototype feature                                       | Day | What we do                                         |
|---------------------------------------------------------|-----|----------------------------------------------------|
| <b>Demand &amp; revenue engine</b>                      | 6   | Reconnect, per-date rates, forecast and pricing UI |
| <b>Action queue<br/>(approve/adjust/snooze/dismiss)</b> | 6   | Full card flow with drivers and confidence         |
| <b>Undo</b>                                             | 6   | Safe revert plus countdown toast                   |
| <b>Predictive maintenance</b>                           | 7   | BMS + QR data, risk and sensor screens             |
| <b>Workforce optimizer</b>                              | 7   | Attendance-driven roster and gap warnings          |
| <b>Purchase suggestions</b>                             | 5-6 | Driven by real inventory levels                    |
| <b>Guest intelligence &amp; Guest DNA</b>               | 8   | Real sentiment model, profile cards                |
| <b>AI concierge</b>                                     | 8   | Chat UI for guests and staff escalation            |
| <b>Notifications outbox</b>                             | 9   | Outbox page with retry                             |
| <b>Feedback loop / learning</b>                         | 9   | Learning page with accuracy and confidence         |
| <b>Revenue simulator</b>                                | 9   | Slider UI on live data                             |
| <b>Cold start readiness</b>                             | 9   | Per-engine readiness banner                        |
| <b>Shadow mode</b>                                      | 9   | Global toggle and preview banner                   |
| <b>CSV data import</b>                                  | 9   | Upload with dry-run results                        |
| <b>3D resort twin</b>                                   | 9   | Lazy-loaded; 2D fallback on mobile                 |
| <b>Roles &amp; audit log</b>                            | 2   | Full permission matrix                             |
| <b>Live WebSocket updates</b>                           | 3   | Authenticated live events                          |
| <b>Known bugs</b>                                       | 1   | Rates, undo, timezone, sentiment                   |

## How we work

- **Daily 15-min stand-up:** done, doing, blocked.
- **Day 1:** backend shares API contracts so frontend builds with mock data in parallel.
- **Every evening:** merge, deploy to staging, demo the day's outcome.
- If a task slips, cut scope, not quality; move extras to the backlog.

## Backlog (after the 10 days)

- Marketing website, real PMS/BMS integrations, real payments.
- Billing folio, restaurant POS, banquets, security gate log.
- Hindi support, native apps, multi-property.